

Lecture 12

Blade dynamic design

ASDA/Dynamics of Rotor Systems

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This lecture

- Frequency diagrams and their construction
 - Using BVP solver
 - Using Southwell's formula and assumed modes
- Frequency diagrams in design and analysis

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Frequency diagrams

Frequency diagrams show how the blade's natural frequencies change with increasing rotor speed. These curves (natural frequency loci) represent the *dynamics* of elastic rotating blades. *Harmonic lines* are used in these diagrams to indicate the rotor excitation frequencies at 1Ω , 2Ω , 3Ω , 4Ω , etc.

Inputs: aero loads, ...

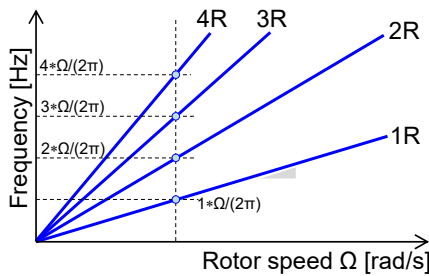
Blade

Outputs: blade deflections ...

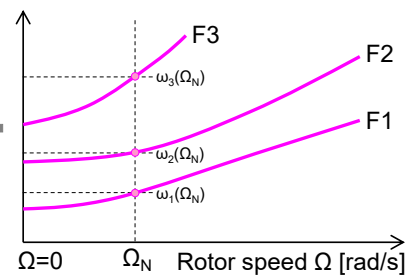
While we might not know the exact rotor-induced loads, we know their excitation frequencies since they are closely linked to the main rotor speed.

The blade can encounter resonant excitation or other destructive in-flight interactions.

Harmonic lines (excitation frequencies):



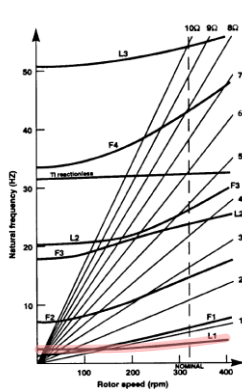
Frequency loci (blade dynamics):



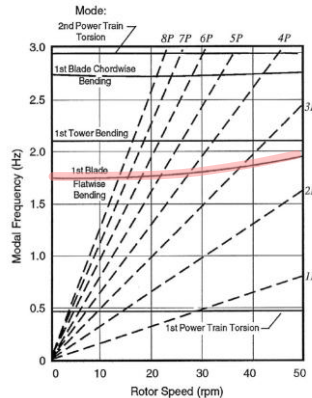
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Example frequency diagrams

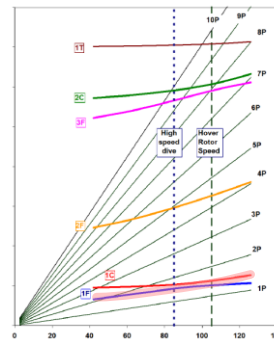
Depending on the engineering discipline or application, frequency diagrams are also known by various other names, such as Campbell diagrams, Southwell diagrams, or fan diagrams.



Lau et al., Performance and Rotor Loads Measurements of the Lynx XZ170 Helicopter with Rectangular Blades, NASA Memorandum 104000



Thresher et al., Structural Dynamic Behavior of Wind Turbines, NASA Technical Report

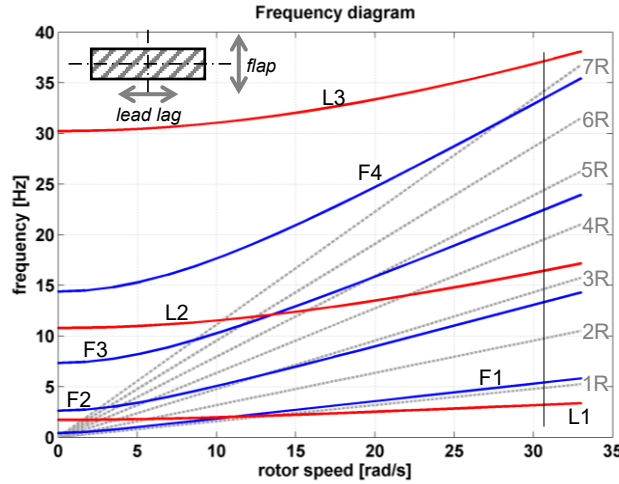


Walsh et al., High Airspeed Testing of the Sikorsky X2 Technology TM Demonstrator, 67th AHS Forum, 2011

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Frequency diagram constructed using BVP4C

We can use an eigenvalue BVP solver (i.e., BVP4C) to compute the eigenvalue (natural frequency) loci for a range of flapping and lead-lag modes. The parameter to be varied in is the rotor frequency Ω .



Note the characteristic trends of F1 and L1 and compare them with the similar rigid blade representations. Further, note various intersection points between the lines and eigenvalue loci.

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The Southwell's formula

An alternative to using BVP solvers is to apply *analytical* methods such as the *Galerkin's method*. The field equation is integrated along the blade's span, and the resulting equation is used to derive a formula for the natural frequency variation, $\omega_F = f(\Omega)$. In the rotorcraft community, this relation is sometimes referred to as the *Southwell's formula*.

Consider the flapping blade eigenvalue problem written for the j -th solution pair:

$$(EI_F w_j'')' - (T(x) w_j')' - \omega_{F,j}^2 m w_j = 0, \quad T(x) = \Omega^2 \int_x^R m(x_1) x_1 dx_1 = \Omega^2 \tilde{T}(x)$$

Multiply the equation by the same w_j and then integrate between $e_F R$ and R :

$$\int_{e_F R}^R (EI_F w_j'')' w_j dx - \Omega^2 \int_{e_F R}^R (\tilde{T} w_j')' w_j dx - \omega_{F,j}^2 \int_{e_F R}^R m w_j^2 dx = 0$$

Rearrange the above equation to determine $\omega_{F,j}^2$:

$$\omega_{F,j}^2 = \frac{1}{M_j} \left(\int_{e_F R}^R (EI_F w_j'')' w_j dx - \Omega^2 \int_{e_F R}^R (\tilde{T} w_j')' w_j dx \right), \quad M_j = \int_{e_F R}^R m w_j^2 dx$$

If EI_F , m and w_j are known, this formula allows calculation of $\omega_{F,j}$. w_j is the mode shape of the rotating blade. We assume that this shape does not change with Ω significantly. Based on this assumption, we can use the mode shape for $\Omega=0$ for all considered rotor speeds.

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The Southwell's formula – flapping motion

To progress the derivation further, the individual integrals are integrated *per partes* (integration by parts)

$$\int_a^b f'(x) g(x) dx = [f'(x) g(x)]_a^b - \int_a^b f(x) g'(x) dx$$

Considering the classical hub and tip boundary conditions, the following is written:

$$\begin{aligned} \int_{e_F R}^R (EI_F w_j'')'' w_j dx &= \left[(EI_F w_j'')' w_j \right]_{e_F R}^R - \left[EI_F w_j'' w_j' \right]_{e_F R}^R + \int_{e_F R}^R EI_F w_j'' w_j'' dx \\ &\quad \text{Zero boundary conditions} \\ \int_{e_F R}^R (\tilde{T} w_j')' w_j dx &= \left[\tilde{T} w_j' w_j \right]_{e_F R}^R - \int_{e_F R}^R \tilde{T} w_j' w_j' dx \\ &\quad \text{Zero } T(R) \text{ and root boundary conditions} \end{aligned}$$

The Southwell's formula for the flapping natural frequency of the rotating elastic blade is:

$$\omega_{F,j}^2 = \frac{\int_{e_F R}^R EI_F w_j''^2 dx}{\int_{e_F R}^R m w_j^2 dx} + \Omega^2 \frac{\int_{e_F R}^R \tilde{T} w_j'^2 dx}{\int_{e_F R}^R m w_j^2 dx} = \omega_{0,F,j}^2 + \Omega^2 K_{F,j}$$

Non-rotating ω_F^2 Change of ω_F^2 due to centrifugal stiffening The Southwell's formula coefficients

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The Southwell's formula – lead-lag motion

Similar approach is now applied to the lead-lag motion of a flexible rotating blade:

$$\begin{aligned} (EI_L v_j'')'' - \Omega^2 (\tilde{T} v_j')' - (\omega_{L,j}^2 + \Omega^2) m v_j &= 0 \\ \int_{e_L R}^R (EI_L v_j'')'' v_j dx - \Omega^2 \int_{e_L R}^R (\tilde{T} v_j')' v_j dx - (\omega_{L,j}^2 + \Omega^2) \int_{e_L R}^R m v_j^2 dx &= 0 \\ \omega_{L,j}^2 = \frac{1}{M_j} \left(\int_{e_L R}^R (EI_L v_j'')'' v_j dx - \Omega^2 \int_{e_L R}^R (\tilde{T} v_j')' v_j dx - \Omega^2 M_j \right), \quad M_j = \int_{e_L R}^R m v_j^2 dx \end{aligned}$$

The Southwell's formula for the lead-lag natural frequency of the rotating elastic blade:

$$\omega_{L,j}^2 = \frac{\int_{e_L R}^R EI_L v_j''^2 dx}{\int_{e_L R}^R m v_j^2 dx} + \Omega^2 \frac{\int_{e_L R}^R \tilde{T} v_j'^2 dx - \int_{e_L R}^R m v_j^2 dx}{\int_{e_L R}^R m v_j^2 dx} = \omega_{0,L,j}^2 + \Omega^2 (K_{L,j} - 1)$$

Non-rotating ω_F^2 Change of ω_F^2 due to centrifugal stiffening The Southwell's formula coefficients

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Assumed and real mode shapes

The Southwell's formulas were derived from the original BVPs. This fact implies that w_j and v_j should be the actual mode shapes calculated for each given Ω . These shapes, however, are generally not known. Due to their similarity with the non-rotating mode shapes ($\Omega=0$), these modes can be used in the formulas instead, with only a small error in the results.

Historically, even the non-rotating modes at $\Omega=0$ were not generally available, and various **approximation strategies** were therefore developed. One approach uses the *assumed mode shapes*. These are chosen to be simple functions $f(x)$ that are similar to the required mode shapes, and which satisfy the problem's BCs as well as, possibly, some other conditions.

$w_j(x) \approx f(x)$	$v_j(x) \approx f(x)$
$\omega_{F,j}^2 \approx \frac{\int_{e_F R}^R EI_F f''^2 dx}{\int_{e_F R}^R m f^2 dx} + \Omega^2 \frac{\int_{e_F R}^R \tilde{T} f'^2 dx}{\int_{e_F R}^R m f^2 dx}$	$\omega_{L,j}^2 \approx \frac{\int_{e_L R}^R EI_L f''^2 dx}{\int_{e_L R}^R m f^2 dx} + \Omega^2 \frac{\int_{e_L R}^R \tilde{T} f_j'^2 dx - \int_{e_L R}^R m f^2 dx}{\int_{e_L R}^R m f^2 dx}$

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Example A: Polynomial assumed mode shape

Consider the case where a *quartic polynomial* is used to construct the *assumed mode shape* function $f(x)$ for F1 mode of a non-rotating elastic blade with zero flapping hinge offset. Assume the cantilevered root and free blade tip.

General *quartic polynomial* $p(x)$, $0 \leq x \leq R$:

$$p(x) = ax^4 + bx^3 + cx^2 + dx + e \quad a, b, c, d, e = ?$$

Boundary and **normalization conditions**, $x=0$ & $x=R$: $f(x) = p(x)$

$$\begin{aligned} x=0: f(x) &= 0 \\ x=0: f'(x) &= 0 \\ x=R: M(x) &= 0 \Rightarrow EI f''(x) = 0 \\ x=R: S(x) &= 0 \Rightarrow EI f'''(x) = 0 \\ x=R: f(x) &= 1 \end{aligned} \Rightarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 12R^2 & 6R & 2 & 0 & 0 \\ 24R & 6 & 0 & 0 & 0 \\ R^4 & R^3 & R^2 & R & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \\ e \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Solve for the polynomial coefficients and construct the final $f(x)$:

$$\mathbf{Ax} = \mathbf{b} \Rightarrow a, b, c, d, e = \dots \Rightarrow f(x) = \frac{1}{3} \left(\frac{x}{R} \right)^4 - \frac{4}{3} \left(\frac{x}{R} \right)^3 + 2 \left(\frac{x}{R} \right)^2$$

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Example A: Polynomial assumed mode shape, cont.

The Southwell's formula with the assumed F1 mode $f(x)$:

$$\omega_{F,1}^2 = \frac{\int_0^R EI_F w_1''^2 dx}{\int_0^R m w_1^2 dx} + \Omega^2 \frac{\int_0^R \tilde{T} w_1'^2 dx}{\int_0^R m w_1^2 dx} \approx \frac{EI \int_0^R \underline{f''^2} dx}{m \int_0^R \underline{f^2} dx} + \Omega^2 \frac{\frac{m}{2} \int_0^R (R^2 - x^2) \underline{f'^2} dx}{m \int_0^R \underline{f^2} dx}$$

Calculations involving the above Southwell's formula with $f(x)$ are straightforward, however, the algebra might be relatively lengthy.

Assume the blade where $EI=5469 \text{ Nm}^2$, $m=4.3 \text{ kg/m}$ and $R=5 \text{ m}$:

$$EI \int_0^R f''^2 dx = 140.006 \text{ kg s}^{-2}, \quad \frac{m}{2} \int_0^R (R^2 - x^2) f'^2 dx = 6.477 \text{ kg}, \quad m \int_0^R f^2 dx = 5.521 \text{ kg}$$

The resulting Southwell's formula expression is:

$$\omega_{F,1}^2 \approx \frac{140.006}{5.521} + \Omega^2 \frac{6.477}{5.521} = (5.036 \text{ rad/s})^2 + 1.173 * \Omega^2$$

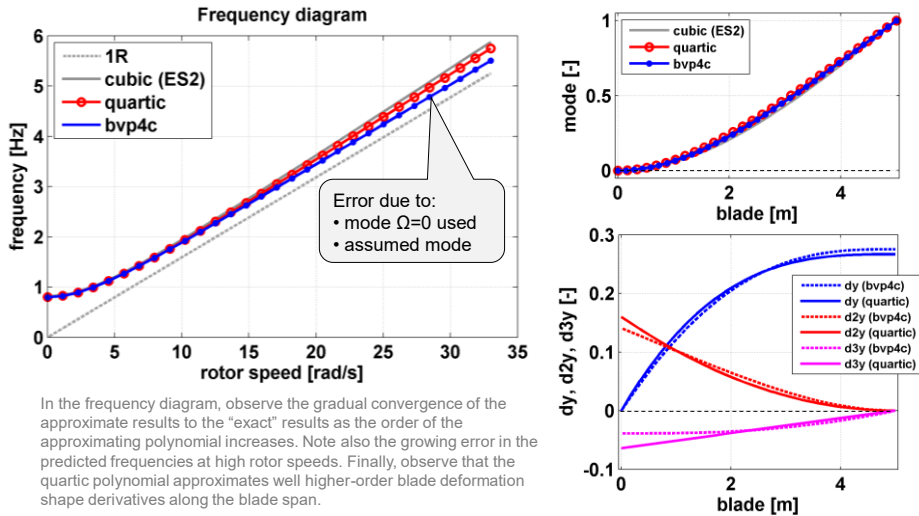
Southwell's coefficients: $\omega_{0,F,1} = 5.036 \text{ rad/s}$, $K_{F,1} = 1.173$

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Example B: Approximate frequency diagrams

Previous approximate results, based on the quartic polynomial *assumed mode-shape* function $f(x)$, are compared here with the "exact" *bvp4c* results and a coarser cubic assumed mode-shape approximation.



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Analysis of the Southwell's formulas

The Southwell formulas allow interpretation of changes in blade dynamics during rotation. Two special cases are considered: the non-rotating condition and the asymptotic case corresponding to very high rotor speed.

$$\omega_{F,j}^2 = \omega_{0,F,j}^2 + \Omega^2 K_{F,j}$$

$$\omega_{F,j} = \sqrt{\omega_{0,F,j}^2 + \Omega^2 K_{F,j}}$$

$$\omega_{L,j}^2 = \omega_{0,L,j}^2 + \Omega^2 (K_{L,j} - 1)$$

$$\omega_{L,j} = \sqrt{\omega_{0,L,j}^2 + \Omega^2 (K_{L,j} - 1)}$$

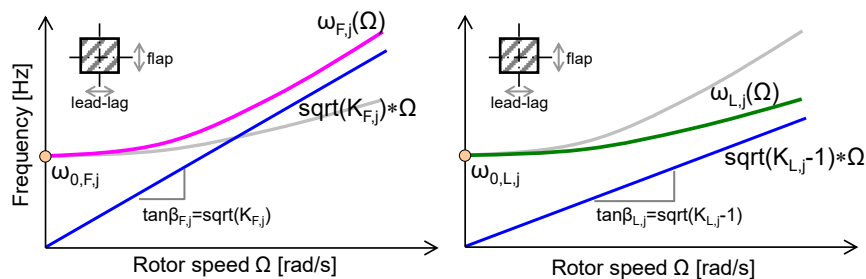
Consider the two special (limiting) cases:

$$\Omega = 0: \omega_{F,j} = \sqrt{\omega_{0,F,j}^2}$$

$$\Omega \gg 0: \omega_{F,j} = \Omega \sqrt{K_{F,j}}$$

$$\Omega = 0: \omega_{L,j} = \sqrt{\omega_{0,L,j}^2}$$

$$\Omega \gg 0: \omega_{L,j} = \Omega \sqrt{(K_{L,j} - 1)}$$

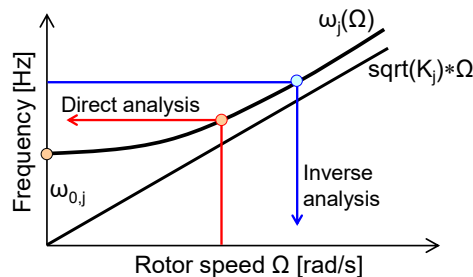


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Frequency diagrams in applications

We distinguish two basic ways of using the Southwell's formulas:

- **Direct analysis:** for provided Ω find the corresponding ω_j (e.g., design studies)
- **Inverse analysis:** for provided ω_j find the corresponding Ω (e.g., freq. placement)



Consider, for instance, lead lag blade dynamics:

Direct analysis:

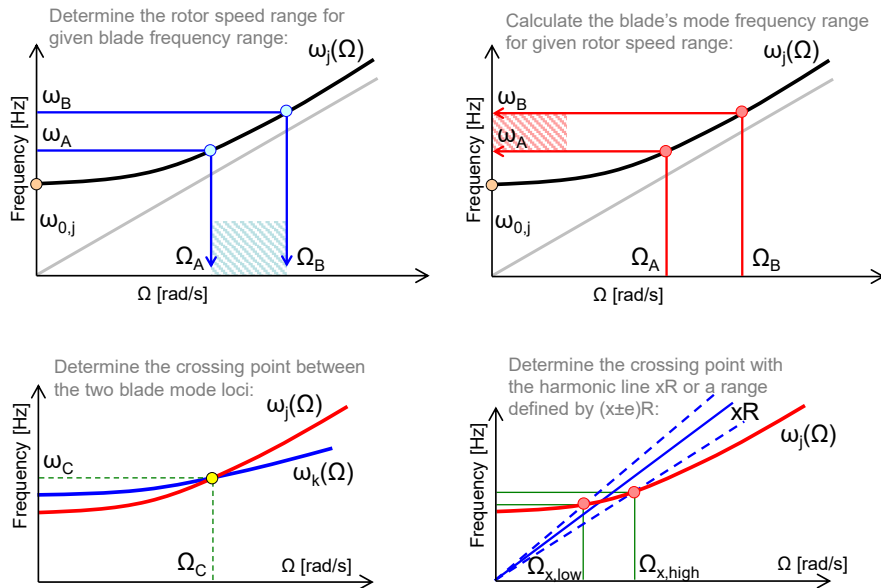
$$\omega_{L,j} = \sqrt{\omega_{0,L,j}^2 + \Omega^2 (K_{L,j} - 1)}$$

Inverse analysis:

$$\Omega = \left(\frac{\omega_{L,j}^2 - \omega_{0,L,j}^2}{K_{L,j} - 1} \right)^{1/2}$$

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Frequency diagrams in applications, cont.



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Summary

- Frequency diagrams show changes and relationships between the rotor-induced frequencies and blade natural frequencies.
- The Southwell's formulas offer direct insights into the blade frequency changes and their driving factors.
- The assumed mode shapes and Southwell's formulas offer simple ways to predict the blade frequency changes.
- Blade frequency diagram is an important tool used for blade dynamic design and analysis.

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